

Task 26: Analyze Bias and Variance Across Different Models

Simple Linear Regression

Analysis:

Simple Linear Regression uses only one independent variable (House Area) to predict house prices. Due to its simplicity, it generally has **high bias** and **low variance**. It may fail to capture all the factors affecting house prices, leading to **underfitting**.

Multiple Linear Regression

Analysis:

Multiple Linear Regression uses several relevant features such as area, bedrooms, bathrooms, location score, and age of the property. This reduces bias while maintaining a reasonable level of variance, resulting in better prediction accuracy than Simple Linear Regression.

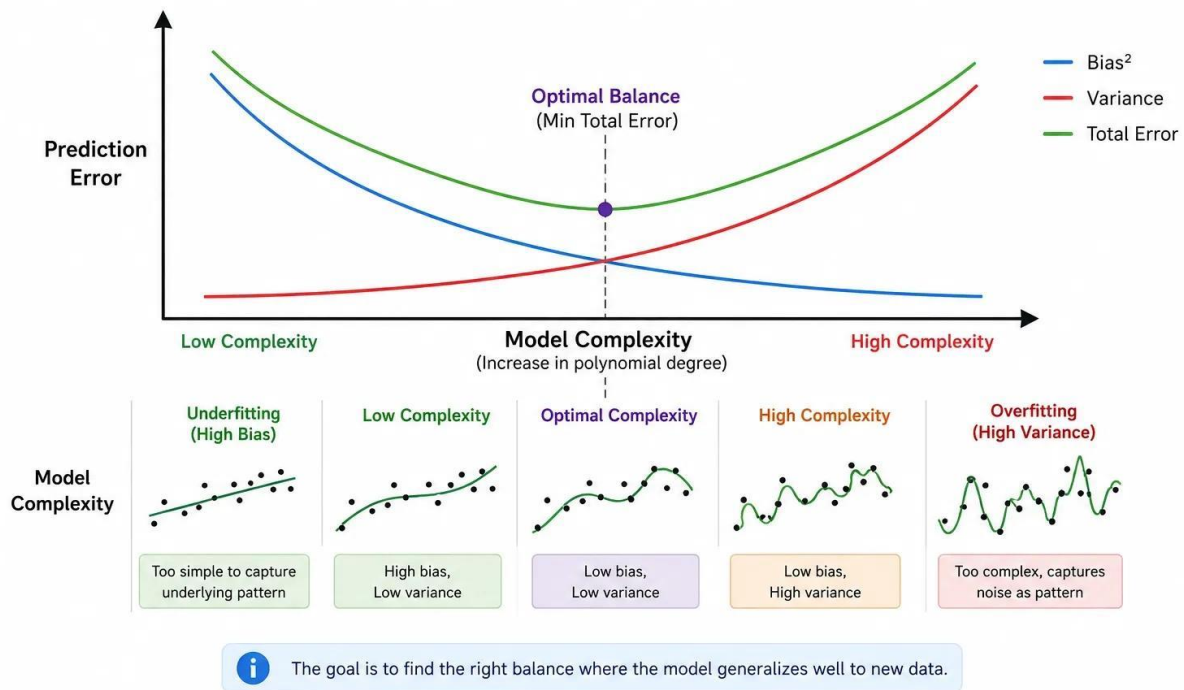
Polynomial Regression

Analysis:

Polynomial Regression can model complex, non-linear relationships between variables. However, increasing the polynomial degree makes the model more complex, which may reduce bias but increase variance. A high-degree polynomial may fit the training data too closely, causing **overfitting**.

Bias-Variance Tradeoff

As model complexity increases, **bias** decreases but **variance** increases.

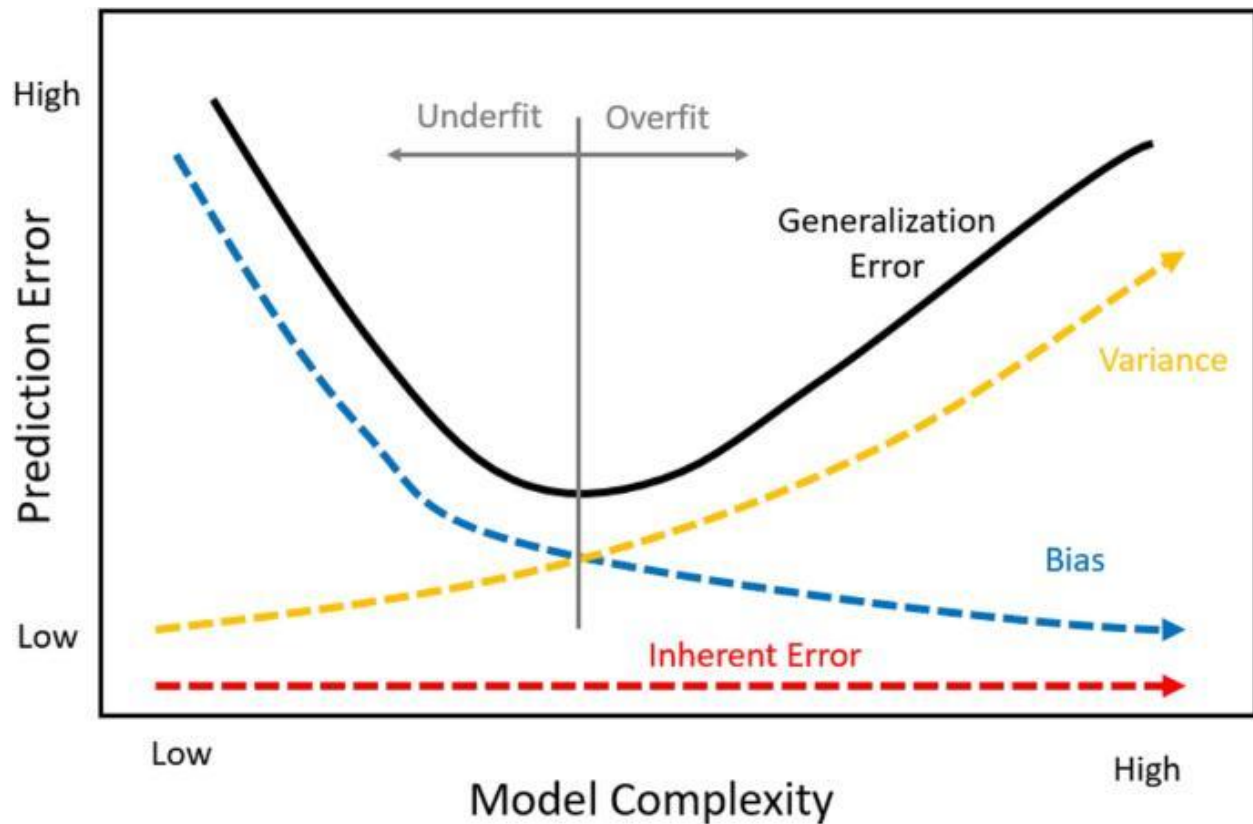


Task 27: Explain How Model Complexity Affects Prediction Error

Answer

Model complexity has a significant impact on prediction error.

- A **simple model** (such as Simple Linear Regression) has **high bias** and **low variance**, which may result in **underfitting** because it cannot capture complex relationships in the data.
- A **highly complex model** (such as high-degree Polynomial Regression) has **low bias** but **high variance**, making it prone to **overfitting** by learning noise in the training data.
- A **moderately complex model** (such as Multiple Linear Regression or a low-degree Polynomial Regression) usually provides the best balance, resulting in better generalization and lower prediction error on unseen data.



Task 28: Identify Which Model Best Balances Bias and Variance

Answer

Based on the model comparison, **Multiple Linear Regression** provides the best balance between bias and variance for this house price prediction dataset.

It uses multiple relevant features to improve prediction accuracy while avoiding unnecessary complexity. Compared to Simple Linear Regression, it captures more information about the data, reducing bias. Compared to high-degree Polynomial Regression, it is less likely to overfit, resulting in better performance on unseen data.

Therefore, **Multiple Linear Regression** is the most suitable model for accurately predicting house prices in this project.

